Cell Death Training App

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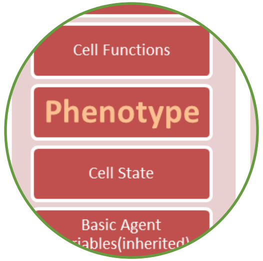
Context



Phenotype

- Motility
- Cell Cycle
- Death
- Mechanics
- Geometry
- Molecular
- Volume

Context

Microenvironment

- Domain Size
- Voxel Size
- Substrates
- Boundary Conditions

Cells

- Custom Data
- Cell Functions
- Phenotype
- Cell State
- Basic Agent Variables(inherited)
- Cell Definition

Global Variables

- User Parameters
- PhysiCell Constants
- List of cells
- SVG options
- MultiCellDS options
- Default Cell Definition
- Default microenvironment

A note about time steps

- PhysiCell is designed to account for the multiple time scales inherent to these problems, and has 3 time scales:
  - $\Delta t_{\mathrm{diffusion}}$ diffusion, secretion, and uptake (default: 0.01 min)
  - $\Delta t_{\mathrm{mechanics}}$ cell movement (default: 0.1 min)
  - Δt_{cell} phenotype and volume changes (default: 6 min)

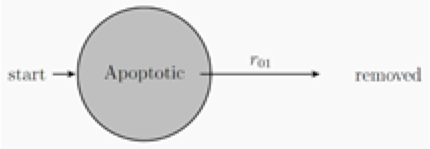
- This allows some efficiency improvements: not all functions need to be evaluated at each time step.

- See the PhysiCell method paper. (Oddly, not in the User Guide (yet).)

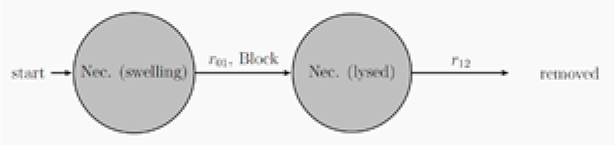
Phenotype: Death

- **Death** has one or more death models:
  - ✦A specialized cycle model with a *removal* transition rate
  - ✦Extra parameters to help govern cell volume
  - ✦Each death model has an associate death rate
  - ✦Also stores an easy Boolean ***dead*** to easily check if the cell is alive.

- PhysiCell has built-in apoptosis and necrosis death models



Apoptosis



Necrosis

Documentation: User Guide, Sec. 11.2

Cell definition: death

```xml
<death>
     <model code="100" name="apoptosis">
          <death_rate units="1/min">0</death_rate>
          <!-- use phase_transition_rates OR phase_durations -->
          <phase_durations units="min">
               <duration index="0" fixed_duration="true">516</duration>
          </phase_durations>
          <parameters>
               <unlysed_fluid_change_rate units="1/min">0.05</unlysed_fluid_change_rate>
               <lysed_fluid_change_rate units="1/min">0</lysed_fluid_change_rate>
               <cytoplasmic_biomass_change_rate units="1/min">1.66667e-02</cytoplasmic_biomass_change_rate>
               <nuclear_biomass_change_rate units="1/min">5.83333e-03</nuclear_biomass_change_rate>
               <calcification_rate units="1/min">0</calcification_rate>
               <relative_rupture_volume units="dimensionless">2.0</relative_rupture_volume>
          </parameters>
     </model>
     <model code="101" name="necrosis">
          <death_rate units="1/min">0.0</death_rate>
          <!-- necrosis uses phase_duration[0] = 0 so that it always immediately
                tries to transition and instead checks volume against the rupture volume -->
          <phase_durations units="min">
               <duration index="0" fixed_duration="true">0</duration>
               <duration index="1" fixed_duration="true">86400</duration>
          </phase_durations>
          <parameters>
               <unlysed_fluid_change_rate units="1/min">0.05</unlysed_fluid_change_rate>
               <lysed_fluid_change_rate units="1/min">0</lysed_fluid_change_rate>
               <cytoplasmic_biomass_change_rate units="1/min">1.66667e-02</cytoplasmic_biomass_change_rate>
               <nuclear_biomass_change_rate units="1/min">5.83333e-03</nuclear_biomass_change_rate>
               <calcification_rate units="1/min">0</calcification_rate>
               <relative_rupture_volume units="dimensionless">2.0</relative_rupture_volume>
          </parameters>
     </model>
</death>
```

- Use death_rate to determine the rate of *starting* each mode of death.
- Use the phase_durations and parameters to control how cells progress through each death model.

Credits

Lesson Planning: Paul Macklin

Slides: Aneequa Sundus, Kali Konstantinopoulos*, Mary Chen*, Furkan Kurtoglu

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Training materials:

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